MATH 475: Honours Partial Differential Equations

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Abstract

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1 Introduction

Textbook used is Rustum Choksi, *Partial Differential Equations, A First Course*, American Mathematical Society. Course grading is divided as follows:

- 10% 5-6 assignments.
- 10% attendance and participation.
- 10% 3-5 quizzes.
- 20% midterm.
- 50% final exam.

2 Background

3 Characteristics

August 31 - September 2, 2026

3.1 Partial Differential Equations

Definition 1 (Partial Differential Equation)**.** *A partial differential equation (PDE) is a differential equation whose unknown function depends on more than one independent variable.*

Remark 1. *Physics rarely hands us a function u directly. It gives a law for how u changes. Example: heat flows from hot to cold, so the temperature $u(x,t)$ in a room changes in time wherever it varies in space. Recovering u from such a relation is the whole subject.*

3.2 Classification

3.2.1 Order

Definition 2 (Order)**.** *The order of a PDE is the order of the highest derivative appearing in it.*

Example. *$u_{xx} = 0$ is second order, and $u_t + uu_x = 0$ is first order.*

3.2.2 Linearity

Definition 3 (Linear PDE)**.** *A PDE is linear if it can be written as $L(u) = f(x)$, where f does not involve u and L satisfies*

$$L(u_1 + c\,u_2) = L(u_1) + c\,L(u_2). \tag{1}$$

Remark 2 (Practical test)**.** *Every term containing u or a derivative of u must satisfy:*

- *u and its derivatives appear only to the first power (no u^2, $(u_x)^2$),*

- *they are never multiplied together (no $u\,u_x$, $u_x u_y$),*
- *they never sit inside a function (no $\sin u$, e^u, $\ln u$),*
- *their coefficients depend only on the independent variables, never on u.*

Each violation creates cross terms, so the superposition identity above fails.

Example. *$u_{xx} + xy\,u_y + e^x u = \cos x$ is linear. Nonlinear: $u\,u_x + u_y = 0$ (product), $u_{xx} + (u_y)^2 = 0$ (power), $u_{xx} + \sin u = 0$ (inside a function).*

Definition 4 (Degrees of nonlinearity)**.** *Let k be the order of the PDE.*

- ***Semilinear:*** *linear in the order-k derivatives, with coefficients depending only on x; nonlinearity only in lower-order terms. Example: $u_{xx} + \sin u = 0$.*
- ***Quasilinear:*** *linear in the order-k derivatives, but their coefficients may depend on u and its lower-order derivatives. Example: Burgers, $u_t + u\,u_x = 0$.*
- ***Fully nonlinear:*** *the order-k derivatives themselves appear nonlinearly. Example: $(u_x)^2 + (u_y)^2 = 1$.*

3.2.3 Homogeneity

Definition 5 (Homogeneous)**.** *A linear PDE $L(u) = f(x)$ is homogeneous if $f \equiv 0$, and inhomogeneous otherwise.*

Example. *$u_t + au_x = x^3$ is linear with $L(u) = u_t + au_x$ and $f(x) = x^3$, so it is inhomogeneous.*

Example. *$u_t + uu_x = f(x)$ is nonlinear, because the term uu_x is not linear in u.*

3.3 Solutions

Definition 6 (Classical solution)**.** *Let the PDE be written as $F(u) = 0$ and let k be its order. A function u is a (classical) solution if*

(1) $u \in C^k$, and

(2) $F(u) = 0$ at every point of the domain.

Example (Transport equation)**.** *Let $u = u(x,t)$ and $F(u_t, u_x) = u_t + cu_x = 0$ with $c \in \mathbb{R}$. Let Φ be any differentiable function and set $u(x,t) = \Phi(x - ct)$. Then*

$$u_x = \Phi'(x - ct), \qquad u_t = -c\,\Phi'(x - ct), \tag{2}$$

so $u_t + cu_x = 0$. The whole family Φ solves the equation. Meaning: the initial profile Φ slides rigidly to the right at speed c.

Example (Heat equation)**.** *Let $u_t = c^2 u_{xx}$ and take the ansatz $u(x,t) = e^{-c^2k^2t}\sin(kx)$ for an arbitrary constant k. Then*

$$u_t = -c^2k^2u, \qquad u_x = k\,e^{-c^2k^2t}\cos(kx), \qquad u_{xx} = -k^2e^{-c^2k^2t}\sin(kx) = -k^2u, \tag{3}$$

so $u_t = c^2u_{xx}$. Meaning: the sine keeps its shape and decays, and high frequencies (large k) decay faster, at rate c^2k^2. Diffusion smooths.

Example (Wave equation)**.** *Let $u_{tt} = c^2u_{xx}$ and take $u(x,t) = \cos(kct)\sin(kx)$. Then*

$$u_{xx} = -k^2u, \qquad u_{tt} = -c^2k^2u, \tag{4}$$

so $u_{tt} = c^2u_{xx}$. Meaning: a standing wave that oscillates forever without decaying. Same spatial profile as the heat example, completely different time behaviour.

3.4 Principle of Superposition

Principle 1 (Superposition)**.** *Let L be linear.*

(1) If $L(u_1) = L(u_2) = 0$, then $L(u_1 + c\,u_2) = 0$ for every $c \in \mathbb{R}$. Solutions of a homogeneous linear PDE form a vector space.

(2) If $L(u_p) = f$ and $L(u_h) = 0$, then $L(u_p + u_h) = f$: one particular solution plus any homogeneous solution solves the inhomogeneous equation.

Proof. By linearity of L,

$$L(u_1{+}cu_2) = L(u_1){+}c\,L(u_2) = 0{+}c{\cdot}0 = 0, \qquad L(u_p{+}u_h) = f{+}0 = f. \ \square \tag{5}$$

Example. *For $u_t = c^2u_{xx}$, both*

$$u_1(x,t) = e^{-c^2k^2t}\sin(kx) \qquad \text{and} \qquad u_2(x,t) = e^{-c^2t}\sin(x) \tag{6}$$

are solutions, so $u(x,t) = a_1u_1(x,t){+}a_2u_2(x,t)$ is a solution for any $a_1, a_2 \in \mathbb{R}$. This is how we will later build series solutions.

[1]

4 General Solutions

4.1 Solving by Direct Integration

Example. *Solve $u_x = 0$ for $u(x,y)$ on $\mathbb{R}^2$. The solution is $u(x,y) = f(y)$ for an arbitrary f.*

[1] 80% chance that what we have written so far is not on the final.

Example. *Solve $u_{xx} = 0$. Set $v(x, y) = u_x$. Then $v_x = 0$, so by the previous example $v(x, y) = f(y)$, hence $u_x = f(y)$. Fixing y and integrating in x (the "constant" of integration only has to be constant in x) gives*

$$u(x, y) = f(y)\, x + g(y). \tag{7}$$

Check: $u_x = f(y)$ and $u_{xx} = 0$.

Example. *Solve $u_{xy} = 0$. Set $v(x, y) = u_x$. Then $v_y = 0$, so $v(x, y) = f(x)$, hence $u_x = f(x)$. Fixing y and integrating in x gives*

$$u(x, y) = F(x) + g(y), \qquad F' = f. \tag{8}$$

Check: $u_x = F'(x)$ depends on x only, so $u_{xy} = 0$.

Remark 3. *In each case the general solution carries arbitrary* functions *rather than arbitrary constants. An ODE of order k has k free constants; a PDE has free functions of the remaining variables. Second order in one variable, or first order in each of two, gives two free functions.*

4.2 Auxiliary Conditions

Auxiliary conditions (initial or boundary conditions) prescribe u, and possibly its derivatives, on a lower dimensional set. They cut the family of general solutions down to one. A typical problem on a domain $\Omega \subset \mathbb{R}^d$ reads

$$\Delta u = f(x) \quad \text{in } \Omega, \qquad u(x) = g(x) \quad \text{on } \partial\Omega. \tag{9}$$

The big three problems of the course:

$$\text{Heat IVP:} \qquad u_t = \alpha u_{xx},\ u(x, 0) = g(x), \tag{10}$$

$$\text{Wave IVP:} \qquad u_{tt} = c^2 u_{xx},\ u(x, 0) = g(x),\ u_t(x, 0) = h(x), \tag{11}$$

$$\text{Dirichlet problem:} \qquad \Delta u = 0 \text{ in the unit disk } D,\ u = g \text{ on } \partial D. \tag{12}$$

Definition 7 (Well-posed, Hadamard)**.** *A PDE with auxiliary conditions is well-posed if*

(1) ***Existence:*** *a solution exists,*

(2) ***Uniqueness:*** *there is only one,*

(3) ***Stability:*** *small changes in the data give small changes in the solution.*

Stability matters because data comes from measurements (with errors) and computers discretize.

Remark 4 (Counting conditions)**.** *For ODEs, number of conditions = order. For PDEs this fails: all three problems above are second order, but the wave IVP needs two conditions while heat and Laplace need one.*

4.3 Operator Viewpoint

Write a linear PDE as

$$Lu = f, \tag{13}$$

where L is the operator, u is the unknown and f is the input. The linear algebra analogue is

$$Mx = b, \qquad x, b \in \mathbb{R}^d, \qquad x = M^{-1}b, \tag{14}$$

so solving a linear PDE is the infinite-dimensional version of inverting M. Well-posedness already shows up here:

- M invertible: exactly one solution for every b.
- M not invertible, b in the range of M: solutions exist but are not unique (add anything in the null space).
- M not invertible, b not in the range: no solution.

Example. $M = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. *For $b = (2, 2)$ every $x = (2 - t, t)$ works: existence without uniqueness. For $b = (1, 2)$ there is no solution: the data is incompatible with the equation.*

4.4 How We Will Solve PDEs

(1) Assume a smooth solution exists.

(2) Work out what the equation and conditions force on it.

(3) In good cases this gives an explicit formula.

(4) Verify the formula really solves the problem. The verification is what makes it a proof.

5 Potential Theory

5.1 Newton's Problem

Place a point mass M at the origin and a test mass m at $x \in \mathbb{R}^3 \setminus \{0\}$. The gravitational force on m has magnitude $GmM/\|x\|^2$ and points back toward the origin:

$$\vec{F}(x) = -\frac{GmM}{\|x\|^2} \cdot \frac{x}{\|x\|} = -\frac{GmM}{\|x\|^3}\, x. \tag{15}$$

5.2 Toolkit

Lemma 1 (Derivative of distance is direction)**.** *Let $r(x) = \|x\|$ with $x = (x_1, x_2, x_3) \in \mathbb{R}^3$. Then for $x \neq 0$,*

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{r}, \qquad \nabla r(x) = \frac{x}{r} = \hat{x}. \tag{16}$$

Proof. Start from $r^2 = x_1^2 + x_2^2 + x_3^2$ and differentiate both sides with respect to x_i:

$$2r\frac{\partial r}{\partial x_i} = 2x_i \quad \Longrightarrow \quad \frac{\partial r}{\partial x_i} = \frac{x_i}{r}. \tag{17}$$

This requires $r \neq 0$, so the identity fails at the origin. □

Remark 5. *∇r is the unit vector pointing radially outward, so $\|\nabla r\| = 1$ everywhere it is defined.*

5.3 Radial Chain Rule

Proposition 1. *Let $f : (0, \infty) \to \mathbb{R}$ be differentiable and set $g(x) = f(r(x)) = f(\|x\|)$. Then for $x \neq 0$,*

$$\nabla g(x) = f'(r)\,\nabla r(x) = f'(r)\,\frac{x}{r}. \tag{18}$$

Proof. Componentwise, by the chain rule,

$$\frac{\partial g}{\partial x_i} = f'(r(x))\frac{\partial r}{\partial x_i} = f'(r)\frac{x_i}{r}. \tag{19}$$

□

So the gradient of any radial function points radially, and f' alone controls the magnitude.

Example (Gradient refresher, non-radial)**.** *For $g(x) = x_1x_2 + x_3^2$,*

$$\nabla g(x) = \left(\frac{\partial g}{\partial x_1}, \frac{\partial g}{\partial x_2}, \frac{\partial g}{\partial x_3}\right) = (x_2,\ x_1,\ 2x_3). \tag{20}$$

5.4 The Laplacian

Everything below works in $\mathbb{R}^d$ for any $d \geq 1$. Write $x = (x_1, \ldots, x_d)$.

First form: sum of second partials

Definition 8 (Laplacian)**.**

$$\Delta = \sum_{i=1}^{d} \frac{\partial^2}{\partial x_i^2}, \qquad \Delta f(x) = \sum_{i=1}^{d} \frac{\partial^2 f}{\partial x_i^2}(x). \tag{21}$$

Example. *Let* $f(x_1, x_2, x_3) = x_1^2 - x_2^2$. *Then*

$$\frac{\partial^2 f}{\partial x_1^2} = 2, \qquad \frac{\partial^2 f}{\partial x_2^2} = -2, \qquad \frac{\partial^2 f}{\partial x_3^2} = 0, \tag{22}$$

so $\Delta f = 2 - 2 + 0 = 0$. *A function with* $\Delta f = 0$ *is called* harmonic.

Second form: divergence of the gradient

Definition 9 (Divergence)**.** *For a vector field* $\vec{v}(x) = (v^1(x), \ldots, v^d(x))$, *where the superscripts label components and are not powers,*

$$\nabla \cdot \vec{v} = \operatorname{div} \vec{v} = \sum_{i=1}^{d} \frac{\partial v^i}{\partial x_i}(x). \tag{23}$$

Applying this to the gradient field $\vec{v} = \nabla f$, whose i-th component is $\partial f / \partial x_i$, gives

$$\nabla \cdot (\nabla f) = \sum_{i=1}^{d} \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_i} \right) = \sum_{i=1}^{d} \frac{\partial^2 f}{\partial x_i^2} = \Delta f. \tag{24}$$

Proposition 2.

$$\Delta = \nabla \cdot \nabla = \operatorname{div} \circ \operatorname{grad}. \tag{25}$$

Remark 6. *This second form is the useful one for radial functions: combining it with the radial chain rule turns* Δ *into an ODE in* r.

5.5 Harmonic Radial Functions

Proposition 3 (Radial Laplacian)**.** *For* $g(x) = f(r)$, $r = \|x\|$ *in* $\mathbb{R}^d$, $x \neq 0$*:*

$$\Delta g = f''(r) + \frac{d-1}{r} f'(r). \tag{26}$$

Solve $\Delta f = 0$ in $d = 3$ with the ansatz $f(r) = r^a$:

$$\Delta f = a(a-1) r^{a-2} + \frac{2}{r}\, a r^{a-1} = a(a+1)\, r^{a-2}, \tag{27}$$

which vanishes for $r > 0$ iff $a = 0$ or $a = -1$. So

$$g(x) = A + \frac{B}{\|x\|}. \tag{28}$$

Remark 7. $1/\|x\|$ *is the Newtonian potential:* $\nabla(1/r) = -x/\|x\|^3$, *the inverse-square field. Harmonic away from the origin only. Its divergence is* 0 *for* $x \neq 0$, *yet its flux through every sphere around the origin is the same nonzero number* (-4π). *So all the "source" sits at one point, and no function can describe it. This is the motivation for distributions (Section 8).*

6 First Order PDEs and the Solution Operator

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6.1 The Underlying ODE

Start from the ODE

$$x'(s) = v(x(s), s), \tag{29}$$

where v is a velocity field: a particle at x at time s moves with velocity $v(x, s)$. Assume $v(x, s)$ is smooth. By the existence and uniqueness theorem there is a unique trajectory through every point, forwards and backwards in time.

6.2 The Solution Operator

Definition 10 (Solution operator)**.** *Given times t_0 and t_1, the solution operator is the map*

$$S_{t_0 \to t_1}(x_0) = x(t_1), \tag{30}$$

where $x(\cdot)$ solves the ODE with initial condition $x(t_0) = x_0$. In words: where is the particle at time t_1 if it was at x_0 at time t_0?

Proposition 4 (Identities)**.** *1. $S_{t \to t}(x) = x$, that is $S_{t \to t} = \mathrm{Id}$.*

2. $S_{t_1 \to t_2} \circ S_{t_0 \to t_1} = S_{t_0 \to t_2}$ (two legs or one, same endpoint).

3. $S_{t_1 \to t_0} = (S_{t_0 \to t_1})^{-1}$: running time backwards inverts the operator.

Example (Constant velocity)**.** *$v = c$. Trajectories are lines $x(t) = x_0 + c(t - t_0)$, so $S_{t_0 \to t_1}(x) = x + c(t_1 - t_0)$.*

Example (Linear velocity)**.** *$v(x) = -x$. Trajectories are $x(t) = x_0 e^{-(t - t_0)}$, so $S_{t_0 \to t_1}(x) = x\, e^{-(t_1 - t_0)}$. In the (x, t) plane these are exponential curves collapsing toward $x = 0$ (the constant velocity ones are parallel lines).*

6.3 Transporting Labels: Where the PDE Comes From

Stamp each particle at time 0 with a label $g(x_0)$ that never changes. Let $u(x, t)$ be the label of the particle sitting at x at time t. That particle started at $S_{t \to 0}(x)$ (run the ODE backwards), so

$$u(x, t) = g\big(S_{t \to 0}(x)\big). \tag{31}$$

Labels are carried forward by the flow; to evaluate u we track backward to the data.

Example. *Constant velocity: $u(x, t) = g(x - ct)$, the signal translated at speed c. Linear velocity $v = -x$: $u(x, t) = g(xe^t)$.*

Deriving the PDE. u is constant along every trajectory, so by the chain rule

$$0 = \frac{d}{dt} u(x(t), t) = u_t + x'(t)\, u_x = u_t + v(x, t)\, u_x. \tag{32}$$

Since a trajectory passes through every point,

$$u_t + v(x,t)\,u_x = 0 \qquad \text{(linear transport equation)}. \tag{33}$$

Remark 8. *We wrote the solution first and got the PDE by differentiating. The method of characteristics is the reverse trip: read v off the PDE, rebuild the flow, then "flow backwards, read the label". The trajectories of v are exactly the characteristics. Formula (31) only defines u where the backward trajectory hits the data line, so the data must meet each trajectory exactly once. Units: v is length per time.*

6.4 Examples of Trajectories

Example. *The curve $\vec{x}(s) = (x(s), y(s)) = (3e^{s-2}, 4e^{s-2})$ satisfies $\vec{x}'(s) = \vec{x}(s)$, so it is a trajectory of the velocity field $\vec{v}(x,y) = (x,y)$. It passes through $\vec{x}(2) = (3,4)$ and $\vec{x}(5) = (3e^3, 4e^3)$.*

Example. *The curve $\vec{x}(s) = (3s+5, 5s+6)$ has constant velocity $\vec{v}(s) = \vec{x}'(s) = (3,5)$, with $\vec{x}(0) = (5,6)$ and $\vec{x}(2) = (11,16)$.*

Example (Decoupled linear field)**.** *Take $\vec{v}(x,y) = (c_1 x, c_2 y)$ for constants c_1, c_2, so the ODE is*

$$\begin{cases} x'(s) = c_1 x(s) \\ y'(s) = c_2 y(s) \end{cases} \qquad \begin{cases} x(0) = x_0 \\ y(0) = y_0. \end{cases} \tag{34}$$

The solution is $x(s) = x_0 e^{c_1 s}$, $y(s) = y_0 e^{c_2 s}$. Check: $x'(s) = c_1 x_0 e^{c_1 s} = c_1 x(s)$, and similarly for y.

Example (Rotation field)**.** *Take $\vec{v}(x,y) = (-y, x)$. A circle of radius r, $\vec{x}(s) = (r\cos s, r\sin s)$, has $\vec{x}'(s) = (-r\sin s, r\cos s) = (-y, x)$, so circles about the origin are trajectories. The general solution of*

$$\begin{cases} x' = -y \\ y' = x \end{cases} \qquad \begin{cases} x(0) = x_0 \\ y(0) = y_0 \end{cases} \tag{35}$$

is

$$x(s) = x_0 \cos s - y_0 \sin s, \qquad y(s) = x_0 \sin s + y_0 \cos s. \tag{36}$$

The circle above is the case $x_0 = r$, $y_0 = 0$.

6.5 Constant Coefficient Transport

Consider

$$a u_x + b u_y = 0, \qquad u = u(x,y), \qquad a, b \in \mathbb{R}. \tag{37}$$

The left side is $\nabla u \cdot (a,b)$, the directional derivative of u in the direction (a,b). So the PDE says u does not change along lines parallel to (a,b).

Proposition 5 (Constancy along characteristics)**.** *Let $\vec{v} = (a, b)$ and let $\vec{x}(s) = (x(s), y(s))$ solve $\vec{x}'(s) = \vec{v}$, that is $x'(s) = a$ and $y'(s) = b$. Then $u(\vec{x}(s))$ is constant in s.*

Proof. By the chain rule,

$$\frac{d}{ds} u(x(s), y(s)) = u_x\, x'(s) + u_y\, y'(s) = a u_x + b u_y = 0. \qquad \square \quad (38)$$

The trajectories of $\vec{v}$ are called *characteristics*. Since u is constant along each one, its value is determined by wherever the characteristic meets the data.

Definition 11 (Characteristic, informal)**.** *A curve along which the PDE reduces to an ODE. Knowing u at one point of it, the PDE determines u along the whole curve. (Special to first order: u at a point depends on the data at a single point. Not true for heat or Laplace.)*

Proposition 6 (General solution)**.** *For any differentiable f,*

$$u(x, y) = f(-bx + ay) \quad (39)$$

solves $a u_x + b u_y = 0$.

Proof. $u_x = -b\, f'(-bx + ay)$ and $u_y = a\, f'(-bx + ay)$, so $a u_x + b u_y = (-ab + ab) f'(-bx + ay) = 0$. $\square$

Remark 9. *The argument $-bx + ay$ is constant exactly along the lines of direction (a, b), which are the characteristics. When $b = 0$ the equation reduces to $a u_x = 0$ and the formula gives $u(x, y) = f(ay)$, a function of y alone, as expected.*

6.6 Adding Data

Example. *Solve*

$$\begin{cases} 3u_x + 2u_y = 0 & \text{(PDE)} \\ u(x, 0) = x^3 & \text{(IC)}, \end{cases} \quad (40)$$

where the data is given on the line $\Gamma = \{(x, y) \mid y = 0\}$.

Method 1 (Via the general solution)**.** *With $a = 3$, $b = 2$, the general solution is $u(x, y) = f(-2x + 3y)$. On $y = 0$ this reads $u(x, 0) = f(-2x) = x^3$. Setting $z = -2x$, so $x = -z/2$, gives $f(z) = -z^3/8$. Therefore*

$$u(x, y) = f(-2x + 3y) = -\frac{(-2x + 3y)^3}{8} = \frac{(2x - 3y)^3}{8}. \quad (41)$$

Check: $u(x, 0) = (2x)^3/8 = x^3$, and $3u_x + 2u_y = \frac{3}{8} \cdot 3(2x - 3y)^2 \cdot 2 + \frac{2}{8} \cdot 3(2x - 3y)^2 \cdot (-3) = 0$.

Method 2 (Via characteristics)**.** *(1) u is constant along curves with $x'(s) = 3$, $y'(s) = 2$.*

(2) *These are the lines* $x(s) = 3s + x_0$, $y(s) = 2s + y_0$.

(3) *Given a target point* (x, y), *find the point* $(x_0, 0)$ *on* Γ *whose characteristic reaches it. From* $y = 2s + 0$ *we get* $s = y/2$, *and then* $x = 3s + x_0$ *gives* $x_0 = x - 3y/2$.

(4) *Since* u *is constant along the characteristic,* $u(x, y) = u(x_0, 0) = x_0^3$:

$$u(x, y) = \left(x - \frac{3y}{2}\right)^3 = \frac{(2x - 3y)^3}{8}. \tag{42}$$

Both methods agree.

6.7 Which Data Curves Work?

A data curve works if it meets every characteristic exactly once (here: the x-axis, the y-axis, $y = x^2 \dots$). It fails if it *is* a characteristic. For $3u_x + 2u_y = 0$, u is forced to be constant on the line $2x - 3y = 0$, so:

- nonconstant data on that line: **no solution**;
- constant data c on that line: **infinitely many** ($u = f(2x - 3y)$ with any f having $f(0) = c$).

Either way the problem is not well-posed.

7 Variable Coefficients and Inhomogeneous Transport

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7.1 Warm-up: Curved Characteristics

Example. $u_x + y\,u_y = 0$. *Now the direction of zero change* $(1, y)$ *varies from point to point. Follow curves tangent to it:* $\frac{dy}{dx} = y$, *so* $y = Ce^x$. *Along such a curve* $\frac{d}{dx}u(x, y(x)) = u_x + yu_y = 0$, *so* u *is constant on each exponential curve. Each curve is labelled by* $C = ye^{-x}$, *so*

$$u(x, y) = f(ye^{-x}). \tag{43}$$

7.2 Problem Setup

Problem 1 (Linear transport with variable coefficients)**.** *Inputs:*

(1) *A curve* $\Gamma \subset \mathbb{R}^2$ *and data* $u_0 : \Gamma \to \mathbb{R}$.

(2) *Coefficients* $a, b : \mathbb{R}^2 \to \mathbb{R}$.

Find $u : \mathbb{R}^2 \to \mathbb{R}$ *such that*

$$\begin{cases} a(x, y)\,u_x + b(x, y)\,u_y = 0 & \text{(PDE)} \\ u(x, y) = u_0(x, y) \quad \forall (x, y) \in \Gamma & \text{(IC).} \end{cases} \tag{44}$$

7.3 Method of Characteristics

Method 3 (Homogeneous case)**.** *(1) Solve the characteristic ODE*

$$\begin{cases} x'(s) = a(x(s), y(s)) \\ y'(s) = b(x(s), y(s)) \end{cases} \qquad \begin{cases} x(0) = x_0 \\ y(0) = y_0 \end{cases} \qquad (x_0, y_0) \in \Gamma. \tag{45}$$

(2) Set $z(s) = u(x(s), y(s))$. By the chain rule

$$z'(s) = u_x\, x' + u_y\, y' = a u_x + b u_y = 0, \tag{46}$$

so z is constant and $z(s) = u_0(x_0, y_0)$.

(3) Given a target (x, y), find s_0 and $(x_0, y_0) \in \Gamma$ with $x(s_0) = x$, $y(s_0) = y$. Then

$$u(x, y) = u_0\big(x_0(x, y),\, y_0(x, y)\big). \tag{47}$$

7.4 Inhomogeneous Case

Now allow a right hand side:

$$a(x, y)\, u_x + b(x, y)\, u_y = c_1(x, y)\, u + c_2(x, y). \tag{48}$$

The characteristics are unchanged, but u is no longer constant along them. With $z(s) = u(x(s), y(s))$,

$$z'(s) = a u_x + b u_y = c_1 z + c_2, \tag{49}$$

so the method becomes a three-dimensional ODE system (the characteristic equations):

$$\begin{cases} x' = a(x, y) \\ y' = b(x, y) \\ z' = c_1(x, y)\, z + c_2(x, y) \end{cases} \qquad \begin{cases} x(0) = x_0 \\ y(0) = y_0 \\ z(0) = u_0(x_0, y_0). \end{cases} \tag{50}$$

Because the PDE is linear, the first two equations do not involve z: solve them first, then the z equation. If $c_1 = c_2 = 0$ we are back to the homogeneous case.

Remark 10 (Vocabulary)**.** *The curves $(x(s), y(s))$ in the plane are the* projected characteristics*. The curve $(x(s), y(s), z(s))$ lies on the solution surface $z = u(x, y)$, and its shadow in the plane is the projected characteristic.*

Method 4 (Recipe)**.** *(1) Solve for $x(s), y(s)$ starting on Γ; one free parameter (e.g. x_0) says where on Γ you started.*

(2) Solve for $z(s)$ with $z(0) =$ data at that starting point.

(3) Given (x, y), solve for s and x_0, substitute into z. The final formula contains only x and y. (This is the backward flow $S_{t\to 0}$ again.)

7.5 Worked Example

Example. *Solve*

$$\begin{cases} -y\,u_x + x\,u_y = u & \textit{(PDE)} \\ u(x,0) = g(x) & \textit{(IC)}, \quad \Gamma = \{(x,0) \mid x > 0\}. \end{cases} \tag{51}$$

Solution 1. *Here* $a = -y$, $b = x$, $c_1 = 1$, $c_2 = 0$.

Step 1: characteristic system.

$$x' = -y, \qquad y' = x, \qquad z' = z, \qquad x(0) = x_0,\ y(0) = 0,\ z(0) = g(x_0). \tag{52}$$

Step 2: solve. *From the rotation field example, the general solution of* $x' = -y$, $y' = x$ *is*

$$x(s) = a_1 \cos s - a_2 \sin s, \qquad y(s) = a_1 \sin s + a_2 \cos s. \tag{53}$$

The initial conditions give $a_1 = x_0$, $a_2 = 0$, *so*

$$x(s) = x_0 \cos s, \qquad y(s) = x_0 \sin s, \qquad z(s) = g(x_0)\, e^{s}. \tag{54}$$

Step 3: invert. *Require* $x(s_0) = x$, $y(s_0) = y$:

$$x = x_0 \cos s_0, \quad y = x_0 \sin s_0 \quad \Longrightarrow \quad x_0 = \sqrt{x^2 + y^2}, \qquad s_0 = \arctan(y/x). \tag{55}$$

Step 4: read off u.

$$u(x,y) = z(s_0) = g\Big(\sqrt{x^2+y^2}\Big)\exp\big(\arctan(y/x)\big). \tag{56}$$

In polar coordinates this is $u = g(r)\, e^{\theta}$, *and the PDE is* $\partial_\theta u = u$, *which checks.*

Remark 11. *The characteristics are circles centred at the origin, so every point except the origin is reached from* Γ. *At the origin both coefficients vanish and the characteristic direction degenerates, so* u *is undefined there. The formula with* arctan *is valid for* $x > 0$*; elsewhere use the polar angle* $\theta \in [0, 2\pi)$ *measured from the positive* x*-axis. Note* u *jumps by a factor* $e^{2\pi}$ *across* Γ *itself when* $g \neq 0$, *since going once around the circle multiplies* z *by* $e^{2\pi}$.

7.6 Time as the Parameter

If one variable is time, data is at $t = 0$, and the coefficient of u_t is 1,

$$u_t + a(x,t)\, u_x = \ldots, \tag{57}$$

then the characteristic equation for t is $t' = 1$, $t(0) = 0$, so $t = s$. Drop s and use t itself as the parameter: characteristics are curves $x(t)$ in the (x,t) plane. This is exactly the setting of the solution operator section.

Example (Variable speed transport)**.**

$$u_t - x\,u_x = 0, \qquad u(x,0) = \frac{1}{1+x^2}. \tag{58}$$

Characteristics: $x'(t) = -x$, $x(0) = x_0$, *so* $x(t) = x_0 e^{-t}$, *and* $z' = 0$ *so* $z = \frac{1}{1+x_0^2}$. *Invert:* $x_0 = xe^t$ *(this is* $S_{t\to 0}$*). Therefore*

$$u(x,t) = \frac{1}{1+x^2e^{2t}}. \tag{59}$$

Every point of the signal moves toward 0 *at speed* $|x|$*, so the blip narrows while its height stays* 1*. Variable speed distorts the shape; constant speed only translates it.*

8 Distributions A: The Delta "Function" Is Not a Function

September 16, 2026

Source: Choksi Ch. 5. Goal: a weaker notion of solving, e.g. $u'' = f$, that allows a right hand side like a point source. The one idea to keep: **pointwise values are the wrong data; integrals against test functions are the right data.**

8.1 Integrable Functions

Definition 12. $f : \mathbb{R} \to \mathbb{R}$ *is* integrable *if* $\int_{-\infty}^{\infty} |f(x)|\,dx < \infty$*. It is* locally integrable *if* $\int_a^b |f(x)|\,dx < \infty$ *for all* $a < b$*.*

Example.

- $f(x) = 1$ *(or any polynomial): locally integrable,* $\int_a^b 1\,dx = b - a$*, but not integrable.*
- *If* f *is continuous and* $f(x) = 0$ *for* $|x| > R$*: integrable.*
- $\frac{1}{1+x^2}$*: integrable (total integral* π*).*
- $\frac{1}{\sqrt{|x|}}$*: locally integrable, since* $\int_{-1}^{1} |x|^{-1/2}dx = 4$ *is finite.*
- $\frac{1}{x}$*: not locally integrable,* $\int_0^1 \frac{dx}{x} = \log x\big|_0^1 = \infty$*.*

Remark 12 (Paradigm)**.** *If* $f(x) = g(x)$ *for all* $x \neq 3$ *but* $f(3) \neq g(3)$*, then* $\int f = \int g$ *over every interval, and* $\int f\phi = \int g\phi$ *for every* ϕ*. Integration cannot see single points, so we treat* f *and* g *as the same function.*

8.2 What We Want from δ_0

Informally, δ_0 is a "point mass" at 0: $\delta_0(x) = 0$ for $x \neq 0$ but $\int_{-\infty}^{\infty} \delta_0(x)\,dx = 1$. No function $\delta_0 : \mathbb{R} \to \mathbb{R}$ can do this (by the paradigm, a function that is 0 except at one point has integral 0). What we really want is: for every continuous f,

$$\text{“}\int_{-\infty}^{\infty} \delta_0(x) f(x)\,dx\text{”} = f(0). \tag{60}$$

8.3 First Appearance: the Derivative of Heaviside

$$H(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases} \tag{61}$$

Pointwise $H'(x) = 0$ for $x \neq 0$ and undefined at 0. Ignoring the point gives $H' = 0$, but then H would be constant, which it is not. The whole jump happens at 0. **The computation that reveals H'.** Fix ϕ smooth, vanishing for large $|x|$, and set

$$F(a) = \int_{-\infty}^{\infty} H(x-a)\phi(x)\,dx = \int_{a}^{\infty} \phi(x)\,dx. \tag{62}$$

- Differentiating under the integral: $F'(a) = -\int H'(x-a)\phi(x)\,dx$. If $H' = 0$ this gives 0.
- Directly (fundamental theorem of calculus): $F'(a) = -\phi(a)$. Not zero.

So whatever H' is, $\int H'(x-a)\phi(x)\,dx = \phi(a)$: it picks out the value of ϕ where the jump sits. That is exactly δ at a. **Same idea with a ramp (in class).** Smooth out the jump:

$$g_\epsilon(x) = \begin{cases} 0, & x < 0 \\ x/\epsilon, & 0 \leq x \leq \epsilon \\ 1, & x \geq \epsilon \end{cases} \qquad g_\epsilon'(x) = \begin{cases} 0, & x < 0 \\ 1/\epsilon, & 0 < x < \epsilon \\ 0, & x > \epsilon \end{cases} \tag{63}$$

Then $\int g_\epsilon'(x) f(x)\,dx = \frac{1}{\epsilon}\int_0^\epsilon f(x)\,dx$, the average of f over $[0, \epsilon]$. If f is continuous near 0, this tends to $f(0)$ as $\epsilon \to 0$. As $g_\epsilon \to H$, the derivatives act like δ_0.

8.4 Second Appearance: Spikes

Define triangles of base $2/n$ and height n (area 1):

$$f_n(x) = \begin{cases} n - n^2|x|, & |x| \leq 1/n \\ 0, & |x| > 1/n. \end{cases} \tag{64}$$

Pointwise, $f_n(x) \to 0$ for $x \neq 0$ and $f_n(0) = n \to \infty$: useless, and many sequences with total mass other than 1 have the same pointwise limit. But for continuous ϕ, $\phi \approx \phi(0)$ on the tiny interval $[-1/n, 1/n]$, so

$$\int f_n \phi \, dx \approx \phi(0) \int f_n \, dx = \phi(0) \quad \Longrightarrow \quad \lim_{n\to\infty} \int f_n \phi \, dx = \phi(0). \tag{65}$$

Example (In class)**.** $\phi = 1$*:* $\int f_n = 1$*.* $\phi = x$*:* $\int f_n x \, dx = 0$ *(f_n even, x odd).* $\phi = x^2$*:*

$$2\int_0^{1/n} (nx^2 - n^2x^3)\, dx = 2\left(\frac{1}{3n^2} - \frac{1}{4n^2}\right) = \frac{1}{6n^2} \to 0. \tag{66}$$

Each limit equals $\phi(0)$ (1, 0, 0), as predicted. This is Dirac's picture: a huge spike of area 1, shape irrelevant.

Remark 13 (Punchline)**.** *δ appears when differentiating H and as the limit of concentrating functions. Both times pointwise values fail and integrating against a generic ϕ succeeds. So we will* define *objects by what they do to test functions.*

9 Distributions B: Distributions and Their Derivatives

September 21, 2026

Source: Choksi 5.3, 5.4.

9.1 Motivation: Weak Solutions of $u'' = f$

A classical solution of $u''(x) = f(x)$ needs $u \in C^2$, so f must be continuous. For example $u'' = -1$ on $(-1, 1)$ with $u(-1) = u(1) = 0$ has $u(x) = \frac{1-x^2}{2}$. We want to allow a point source: replace f by concentrating f_ϵ, solve

$$u_\epsilon'' = f_\epsilon, \qquad u_\epsilon(-1) = u_\epsilon(1) = 0, \tag{67}$$

and let $\epsilon \to 0$. To make sense of the limit equation $u'' = \delta_0$ we need distributions.

9.2 Test Functions

Definition 13 (Support)**.** *The support of a continuous ϕ is the closure of $\{x : \phi(x) \neq 0\}$. ϕ has* compact support *if $\phi = 0$ outside some interval $[-R, R]$.*

Definition 14 (Test functions)**.** *$C_c^\infty(\mathbb{R})$ is the set of infinitely differentiable functions with compact support.*

No polynomial works (a polynomial vanishing on an interval is 0). The standard example is the **bump function**

$$\phi(x) = \begin{cases} e^{-1/(1-x^2)}, & |x| < 1 \\ 0, & |x| \geq 1, \end{cases} \tag{68}$$

with support $[-1,1]$. The only issue is smoothness at $x=\pm 1$: every derivative tends to 0 there because the exponential beats every power (Exercise 5.1).

Building more test functions from one: translates and rescalings $\phi(ax+b)$, sums and scalar multiples, products $\psi\phi$ with ψ any C^∞ function, and most importantly **derivatives**: $\phi \in C_c^\infty(\mathbb{R}) \Rightarrow \phi^{(k)} \in C_c^\infty(\mathbb{R})$ for all k. This is why we insist on C^∞.

9.3 Definition of a Distribution

Definition 15 (Distribution)**.** *A distribution F is a rule assigning to each test function ϕ a real number,*

$$F : C_c^\infty(\mathbb{R}) \to \mathbb{R}, \qquad \phi \mapsto \langle F, \phi\rangle \in \mathbb{R}, \tag{69}$$

which is

- ***linear**: $\langle F, a\phi + b\psi\rangle = a\langle F,\phi\rangle + b\langle F,\psi\rangle$,*
- ***continuous**: $\phi_n \to \phi$ in $C_c^\infty(\mathbb{R})$ implies $\langle F,\phi_n\rangle \to \langle F,\phi\rangle$.*

A map from functions to numbers is called a functional. *$\langle F,\phi\rangle$ is just notation for $F(\phi)$, chosen to look like an integral.*

Definition 16 (Convergence in $C_c^\infty(\mathbb{R})$)**.** *$\phi_n \to \phi$ in $C_c^\infty(\mathbb{R})$ if*

(i) all ϕ_n and ϕ vanish outside one common $[-R,R]$, and

(ii) for every $k \geq 0$, $\max_x |\phi_n^{(k)}(x) - \phi^{(k)}(x)| \to 0$ (every derivative converges uniformly).

Distributions form a vector space: $\langle F+G,\phi\rangle = \langle F,\phi\rangle + \langle G,\phi\rangle$ and $\langle cF,\phi\rangle = c\langle F,\phi\rangle$.

Remark 14 (Rule of the game)**.** *Never ask for the value of a distribution at a point x. Ask for its value on a test function ϕ: think of ϕ as a probe.*

9.4 Functions as Distributions

Proposition 7. *If f is locally integrable, then*

$$\langle F_f,\phi\rangle = \int_{-\infty}^{\infty} f(x)\phi(x)\,dx \tag{70}$$

defines a distribution F_f, the distribution generated by *f.*

Proof. If $\phi = 0$ outside $[-R,R]$, then $\left|\int f\phi\right| \leq \max|\phi| \int_{-R}^{R} |f| < \infty$, so it is well defined (this is exactly why we need local integrability). Linearity is linearity of the integral. Continuity: with common support $[-R,R]$,

$$|\langle F_f,\phi_n\rangle - \langle F_f,\phi\rangle| \leq \max|\phi_n - \phi| \int_{-R}^{R} |f|\,dx \to 0. \qquad \square \tag{71}$$

Example (5.3.1)**.** $f \equiv 1$*:* $\langle F_1, \phi \rangle = \int \phi \, dx$.

Remark 15. *Functions that differ only on a set integration cannot see (measure zero) generate the same distribution; otherwise F_f determines f. Notation: we usually just write f for F_f. "In the sense of distributions" means the statement is about F_f.*

9.5 The Delta Function

Definition 17 (5.3.2)**.** $\langle \delta_0, \phi \rangle = \phi(0)$*, and for* $a \in \mathbb{R}$*,* $\langle \delta_a, \phi \rangle = \phi(a)$.

It is linear, and continuous since $\phi_n \to \phi$ uniformly gives $\phi_n(0) \to \phi(0)$.

- There is no integral in the definition. Never write $\delta_0(x)$ or $\int \delta_0(x)\phi(x)\, dx$.
- It is not a function:

Proposition 8. *There is no locally integrable f with $F_f = \delta_0$.*

Proof. Suppose $\int f\phi = \phi(0)$ for every test function. Take $\phi = e \cdot$ (bump), so $\phi(0) = 1$, $0 \le \phi \le 1$, support $[-1, 1]$. Let $\phi_n(x) = \phi(nx)$: support $[-1/n, 1/n]$, $\phi_n(0) = 1$. Then

$$1 = \phi_n(0) = \int f\phi_n \, dx \le \int_{-1/n}^{1/n} |f| \, dx \to 0, \tag{72}$$

since the integral of a locally integrable function over a shrinking interval goes to 0. Contradiction. □

9.6 Derivative of a Distribution

Motivation. For $f \in C^1$ and a test function ϕ, integrate by parts (boundary terms vanish by compact support):

$$\int f'\phi \, dx = -\int f\phi' \, dx. \tag{73}$$

The derivative moves onto ϕ at the cost of a sign, and the right side makes sense even if f is not differentiable.

Definition 18 (5.4.1)**.**

$$\langle F', \phi \rangle = -\langle F, \phi' \rangle. \tag{74}$$

This is a distribution (ϕ' is a test function; linear; continuous since $\phi_n \to \phi$ implies $\phi_n' \to \phi'$). So **every distribution, hence every locally integrable function, has a derivative**.

Proposition 9 (Consistency, 5.20)**.** *If $f \in C^1$, then $(F_f)' = F_{f'}$: the distributional derivative agrees with the classical one. (Proof: the integration by parts above.)*

Example. $\langle \delta_0', \phi \rangle = -\phi'(0)$. *Higher derivatives, one sign per derivative:*

$$\langle F^{(k)}, \phi \rangle = (-1)^k \langle F, \phi^{(k)} \rangle. \tag{75}$$

Remark 16. *Like the transpose in linear algebra ($\langle Ax, y \rangle = \langle x, A^T y \rangle$): the transpose of $\frac{d}{dx}$ is $-\frac{d}{dx}$. Every operation on distributions will be defined this way.*

9.7 Derivatives of Piecewise Smooth Functions

Example (Heaviside, 5.4.1)**.**

$$\langle H', \phi \rangle = -\int_{-\infty}^{\infty} H\phi'\, dx = -\int_0^{\infty} \phi'\, dx = -\big(0 - \phi(0)\big) = \phi(0), \tag{76}$$

using $\phi(\infty) = 0$. So $H' = \delta_0$ in the sense of distributions. (We never wrote $\int \delta_0(x)\phi(x)dx$.)

Example ($|x|$, 5.4.2)**.** *Split at 0 and integrate by parts on each side:*

$$\langle |x|', \phi \rangle = -\int |x|\phi'\, dx = \int_{-\infty}^{0} x\phi'\, dx - \int_0^{\infty} x\phi'\, dx = -\int_{-\infty}^{0} \phi\, dx + \int_0^{\infty} \phi\, dx, \tag{77}$$

(boundary terms vanish: at 0 because of the factor x, at $\pm\infty$ by compact support). So $|x|' = \operatorname{sgn}(x)$ (-1 for $x < 0$, $+1$ for $x > 0$); the value at 0 is irrelevant because $|x|$ is continuous. Second derivative:

$$\langle \operatorname{sgn}', \phi \rangle = -\int \operatorname{sgn}\, \phi' = \int_{-\infty}^{0} \phi' - \int_0^{\infty} \phi' = \phi(0) + \phi(0) = 2\phi(0), \tag{78}$$

so $|x|'' = 2\delta_0$. The jump of size 2 in sgn *shows up as $2\delta_0$.*

Theorem 2 (Derivative of a function with a jump)**.** *Let f be C^1 except at one point a, with f and f' having one-sided limits at a. Let $\{f'\}$ denote the pointwise derivative for $x \neq a$. Then, in the sense of distributions,*

$$f' = \{f'\} + \big(f(a^+) - f(a^-)\big)\, \delta_a. \tag{79}$$

In particular, if f is continuous at a, then $f' = \{f'\}$.

Proof idea. Split $-\int f\phi'$ at a and integrate by parts on each side. The interior terms give $\int \{f'\}\phi$; the boundary terms at a give $f(a^+)\phi(a) - f(a^-)\phi(a)$. □

Checks: H has jump 1 at 0, $\{H'\} = 0$, so $H' = \delta_0$. $|x|$ has no jump, so $|x|' = \operatorname{sgn}$.

Method 5 (Recipe)**.** *(1) Fix a test function ϕ and write $\langle f', \phi \rangle = -\int f\phi'$.*

(2) Split $\mathbb{R}$ into intervals where f is smooth.

(3) Integrate by parts on each piece. Boundary terms at $\pm\infty$ vanish; boundary terms at a singularity add up to (jump of f) $\times\, \phi(a)$, i.e. a delta.

In class: Choksi Exercises 5.2 and 5.8.

9.8 Fundamental Solution of $\frac{d^2}{dx^2}$

Divide the $|x|$ example by 2:

$$\left(\frac{|x|}{2}\right)'' = \delta_0. \tag{80}$$

A solution of $u'' = \delta_0$ is a *fundamental solution*: the response to a unit source at one point. $u = |x|/2$ is continuous, has a kink at the source, and solves $u'' = 0$ away from it. The solution is two derivatives smoother than the right side.

Example. *Back to the motivation:* $u'' = \delta_0$ *on* $(-1, 1)$ *with* $u(\pm 1) = 0$ *is solved by* $u(x) = \frac{|x|-1}{2}$ *(add the constant* $-\frac{1}{2}$*, which has zero second derivative, to fix the boundary values).*

Remark 17. *This is the 1D version of the point mass from week 1: in* $\mathbb{R}^3$*,* $\Delta \frac{1}{|x|} = 0$ *away from* 0*, and in the sense of distributions* $\Delta \frac{1}{|x|} = -4\pi\delta_0$*. Proved later with Laplace's equation (Choksi Ch. 8).*

10 Misc

September 23, 2026

Review C_c^∞: smooth functions with compact support. Distribution. $F : C_c^\infty \to \mathbb{R}$:

1. linear
2. cts
3. write $< F, \phi >= F(\phi)$
4. $F(c_1\phi_1 + c_2\phi_2) = c_1F(\phi_1) + c_2F(\phi_2)$
5. $\phi_n \to \phi \implies F(\phi_n) \to F(\phi)$.

Uniform converence: Convergence in C_c^∞: uniform converge of $\phi_n \to \phi$ also $\phi_n' \to \phi'$, $\phi_n^{(k)} \to \phi^{(k)}$ for all derivatives. Nuild distribution F_f from integrable function f'. Given f, define F_f by $< F_f, \phi >= \int_{-\infty}^{\infty} f(x)\phi(x)dx$.

Example. $f(x) = 3; < F_3, \phi >= 3\int_{-\infty}^{\infty} phi(x)dx$.

Example. $h(x) = \begin{cases} 1, x \geq 0 \\ 0, x < 0 \end{cases}$ $, F_h = H, < H, \phi >= \int_{-\infty}^{\infty} h(x)\phi(x)dx = \int_0^{\infty} \phi(x)dx.$

Derivative of a distribution, given F distribution. Define, another one, F' by $< F', \phi >= - < F, \phi' >$. NOte if $F = F_f$ fore some f, then $F' = F_{f'}$ when f' exists

September 28, 2026

Review: Integration by parts: $\int_a^b (f\phi)'dx = \int_a^b f'\phi dx + \int_a^b f\phi' dx$, $\int_a^b f'\phi dx = -\int_a^b f\phi' dx + f\phi|_a^b$.

Case: $\phi(a) = \phi(b) = 0$, $\int_a^b f\phi dx = -\int_a^b f\phi' dx$. Recall: f integrande fn. F_f distribution define by $< F_f, \phi >= \int_{-\infty}^{\infty} f(x)\phi(x)dx = \int_{-L}^{L} f(x)\phi(x)dx$.

Definition: F distributions define F' by $< F', \phi >= - < F, \phi' >$ suppose function f differentiable f': $< F_{f'}, \phi >= \int_{-\infty}^{\infty} f'(x)\phi(x)dx$. $< F_{f'}, \phi >= - < F_f, \phi >= -\int_{-\infty}^{\infty} f(x)\phi'(x)dx \overset{intbyparts}{=} +\int_{-\infty}^{\infty} f'(x)\phi(x)dx$. For any function $< F_H, \phi >= \int_{-\infty}^{\infty} H(x)\phi(x)dx$.

Distributions $f(x) = |x|, sgn(x) = \begin{cases} -1, x < 0 \\ 0, x = 0 \\ 1, x > 0 \end{cases}, H(x) = \begin{cases} +1, x \geq 0 \\ 0x < 0 \end{cases}$,

$F'_{|x|} = F_{sgn}$, $F'_H = \delta_0, < \delta_0, \phi >= \phi(s)$.

Quiz Hint: Show from definiiton $H'(x) = \delta_0$ as distributions.

Example: $f(x) = \begin{cases} 2x + 3, x < 0 \\ x^2, x \geq 0 \end{cases} = g(x) - 3H(x), g(x) = \begin{cases} 2x + 3, x < 0 \\ x^2 + 3, x \geq 0 \end{cases}$,

$F'_g = F_{g'}$, $(-3H(x))' = -3H' = -3\delta_0$, $F'_f = F_{g'} - 3\delta_0$, where g continuous and piecewise differentiable

October 19, 2026

Check example 5.4.3 in file distribution B

Final Fourier lecture Reminder: series: sine, cosine, full; interval: (0,l),(0,l),(-l,l); Functions $X_n(x)$: $\sin(\frac{n\pi x}{l}), n \geq 1,\ 1, \cos(\frac{n\pi x}{l}), n \geq 1,\ 1, \sin(\frac{n\pi x}{l}), \cos(\frac{n\pi x}{l})$; $||X_n(x)||^2$: l/2, l,l/2, 2l,l,l; Coefficients: $b_n = 2/l \int_0^l \psi \sin(\frac{n\pi x}{l})dx$, $a_n = 2/l \int_0^l \psi \cos(\frac{n\pi x}{l})dx$, $a_n, = 1/l \int_{-l}^{l} \psi \sin(\frac{n\pi x}{l}) \cos(\frac{n\pi x}{l})dx$; respectively.

A Appendix

B Solutions

C Useful Links